Internetworking

Lecture 9: DHCP and IPv6

Literature:

Forouzan, TCP/IP Protocol Suite: Ch 16, 27

DHCP

Autoconfiguration

- Need a way to simply connect a computer to a new network
  - No manual configuration
- But each computer attached to a TCP/IP network must know the following:
  - Its IP address and subnet mask
  - The IP address of a router (default gateway)
  - The IP address of a name server
  - Extra info: default TTL, time servers, etc.
- We have talked about RARP, to translate MAC → IP addr
  - But, RARP needs direct access to the (ethernet) hardware
  - It only contains an IP address
  - Requires fixed hw addresses
- BOOTP and DHCP solve these problems.
  - Client/Server architecture

BOOTP – Bootstrap Protocol

- BOOTP (RFC 951) is a lot more powerful than RARP
- Client/Server Protocol
- Designed to provide the previously mentioned pieces of information (and more)
- BOOTP sends requests/replies over UDP
  - Easy to write a user space server
  - Client does not need a full TCP/IP stack to run BOOTP
- But BOOTP is not dynamic
  - uses a static binding between MAC and IP addresses

DHCP (RFC 1531)

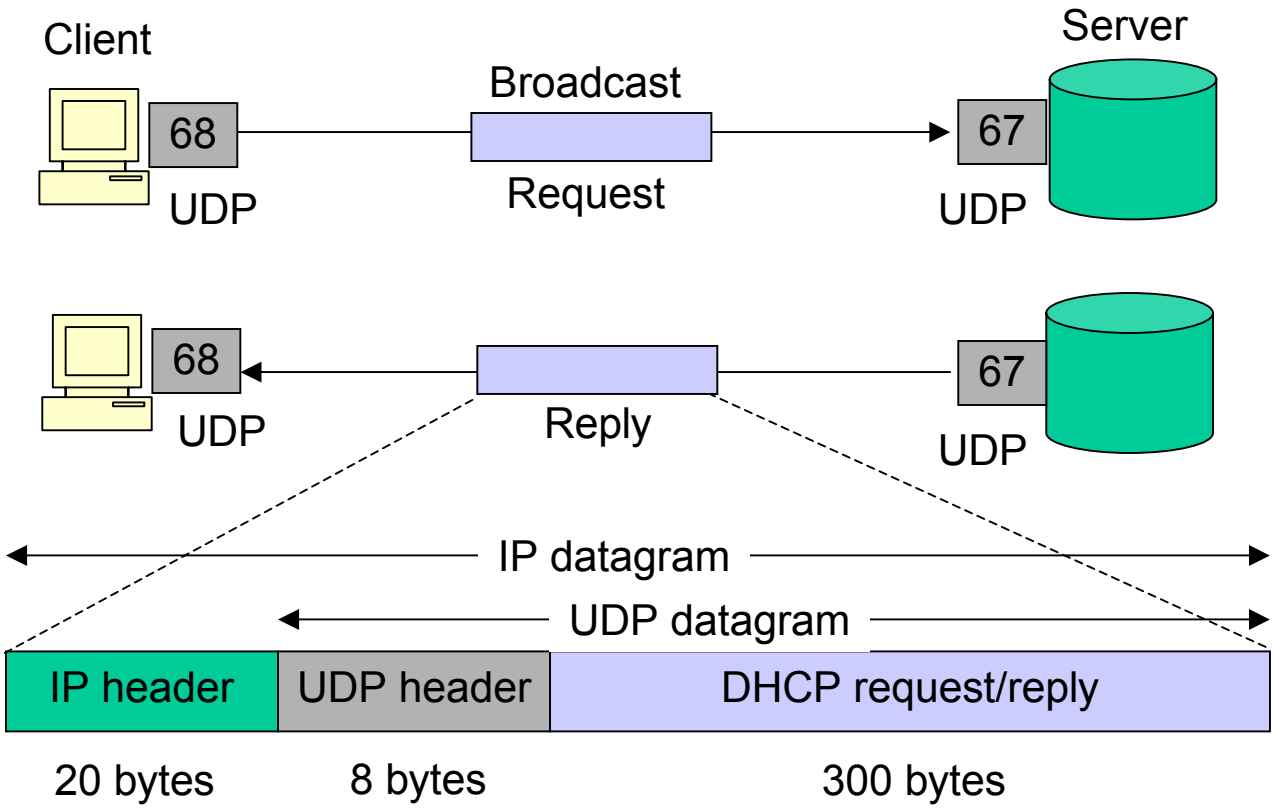
DHCP – Dynamic Host Configuration Protocol

- DHCP provides dynamic configuration
  - Client can get a temporary address, and move from network to network
- DHCP is backward compatible with BOOTP
  - BOOTP client can request a static configuration from a DHCP server

DHCP Operation

1. DHCP server uses UDP port no 67 and waits for client
2. Client sends requests encapsulated in UDP using:
   - dst IP address 255.255.255.255
   - dst port no 67
   - src IP address 0.0.0.0
   - src port no 68
3. Server replies with either broadcast or unicast message using UDP destination port 68
   - The unicast IP address of the client can be used in server's reply message if client's IP address has been manually added in the ARP table
   - Note that the server will know the client's MAC address

DHCP simplified example



Well-known DHCP Port Numbers

Note that DHCP does *not* use *ephemeral* port number at the client side.

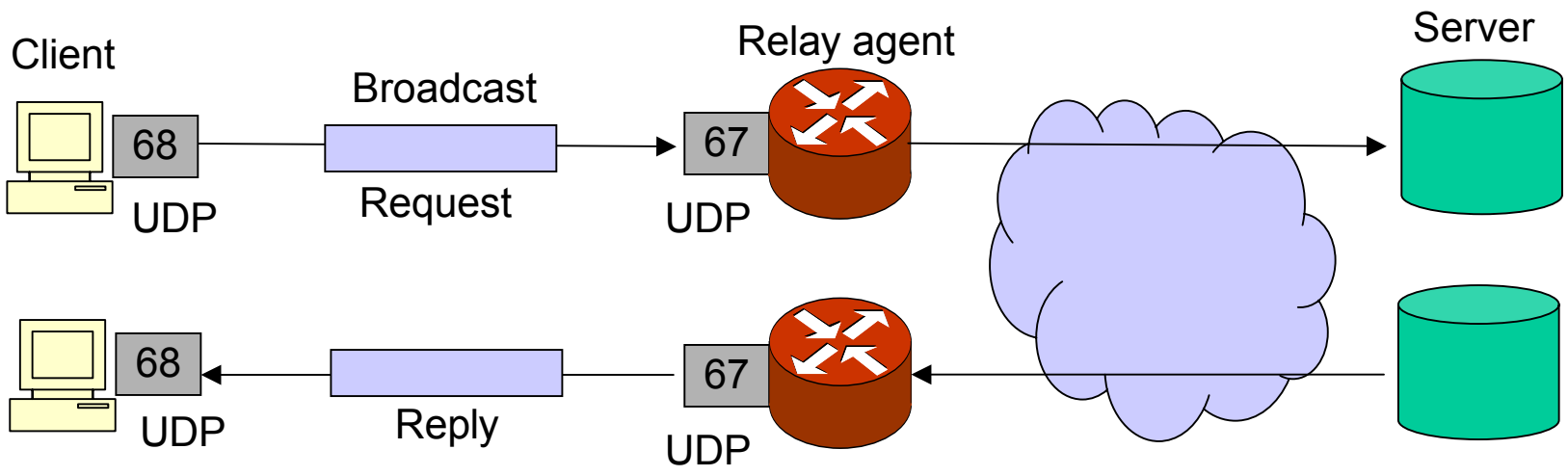
- The reply from the server can be broadcast
- Broadcast response will reach all hosts on the subnet
- It is considered bad manner to broadcast to a random port number

What if two clients are using DHCP simultaneously?

- The Transaction ID will help recognizing the reply message
- Transaction ID is chosen randomly

Relay Agent

- A *relay agent* (proxy) is used so a DHCP server can serve several subnets
  - A relay agent is a router that sends local requests to a remote server, and relays replies back to the subnet.



DHCP Message Header

<table>
<tr><th>0 – 7</th><th>8 – 15</th><th>16 – 23</th><th>24 – 32</th></tr>
<tr><td>Operation Code</td><td>Hardware Type</td><td>Hardware Length</td><td>Hop Count</td></tr>
<tr><td colspan="4">Transaction ID</td></tr>
<tr><td colspan="2">Number of Seconds</td><td>B</td><td>Unused</td></tr>
<tr><td colspan="4">Client IP Address</td></tr>
<tr><td colspan="4">Your IP Address</td></tr>
<tr><td colspan="4">Server IP Address</td></tr>
<tr><td colspan="4">Gateway IP Address</td></tr>
<tr><td colspan="4">Client Hardware Address (16 bytes)</td></tr>
<tr><td colspan="4">Server Name (64 bytes)</td></tr>
<tr><td colspan="4">Boot File Name (128 bytes)</td></tr>
<tr><td colspan="4">Options (variable)</td></tr>
</table>

DHCP Message fields

• Opcode:	Request (1), Reply (2)
• Hardware type:	1 for Ehernet (cf ARP)
• Hardware address len:	6 for Ethernet
• Hop Count	+1 for each relay forwarder (proxy)
• Transaction ID	set by client to identify session
• B-flag	forced broadcast reply
• # seconds	time since boottime of client
• client IP address	If client knows its address
• your IP address	Client address set by server
• server IP address	Server address filled in by server
• gateway IP address	Proxy server address
• client hardware address	Set by client (same as SA in eth header)
• server hostname	Set by server
• boot filename	Set by server for bootstrapping
• options	Variable length (up to 312 bytes)

DHCP Options and message types

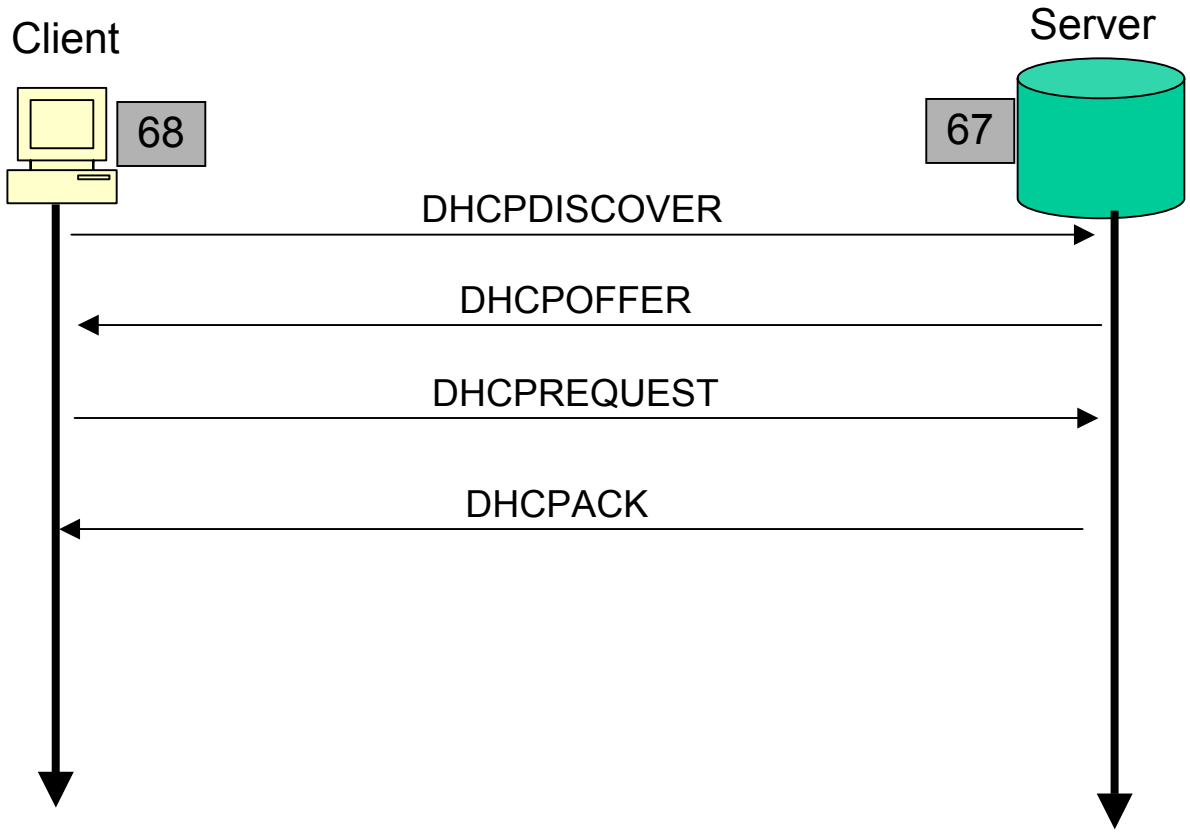
- Lots of semantics implemented in the DHCP options field
  - or ”Vendor-specific field”
- Uses ”TLV”: Type-Length-Value

type	length	value

- Subnetmask
- Address of routers, nameservers, timeservers, hostname, etc.
- Message type examples
  - DHCPDISCOVER
  - DHCPOFFER
  - DHCPREQUEST
  - DHCPACK
  - DHCPNACK
  - DHCPRELEASE

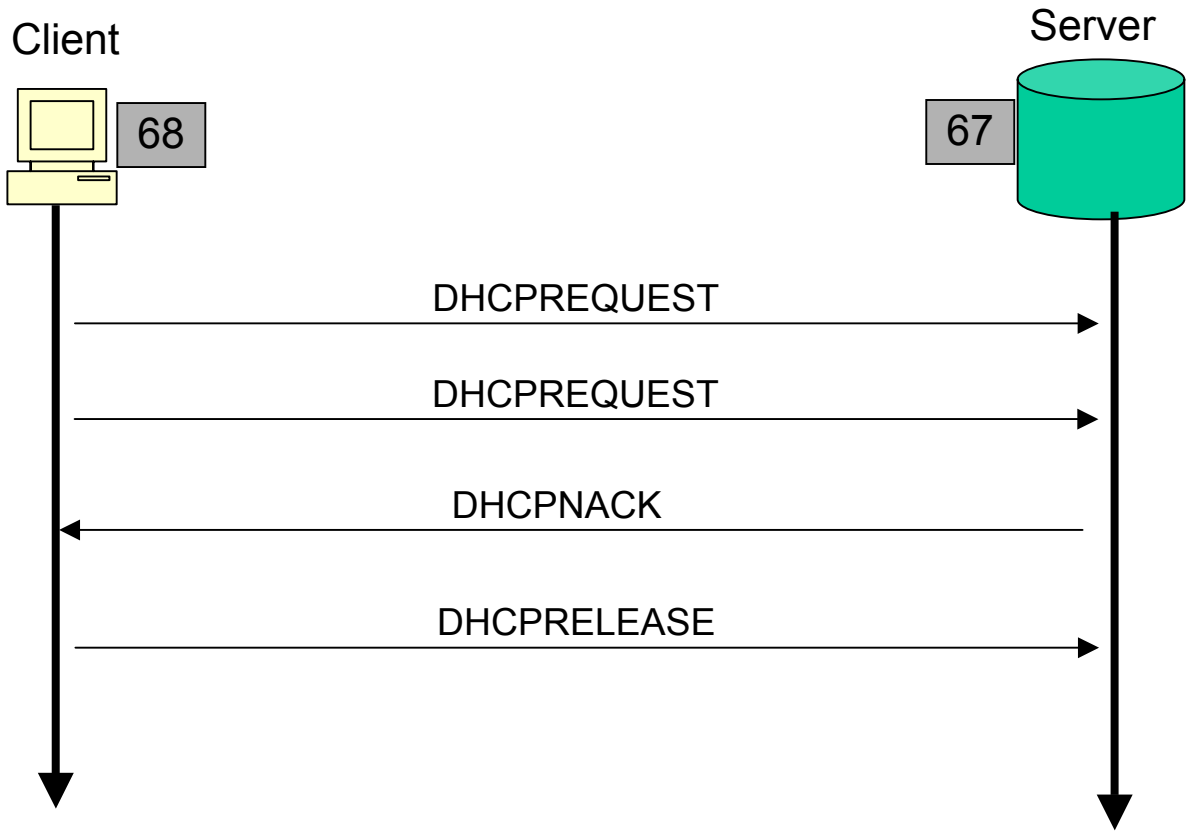
DHCP Initialization

- First discover a server, then request address lease



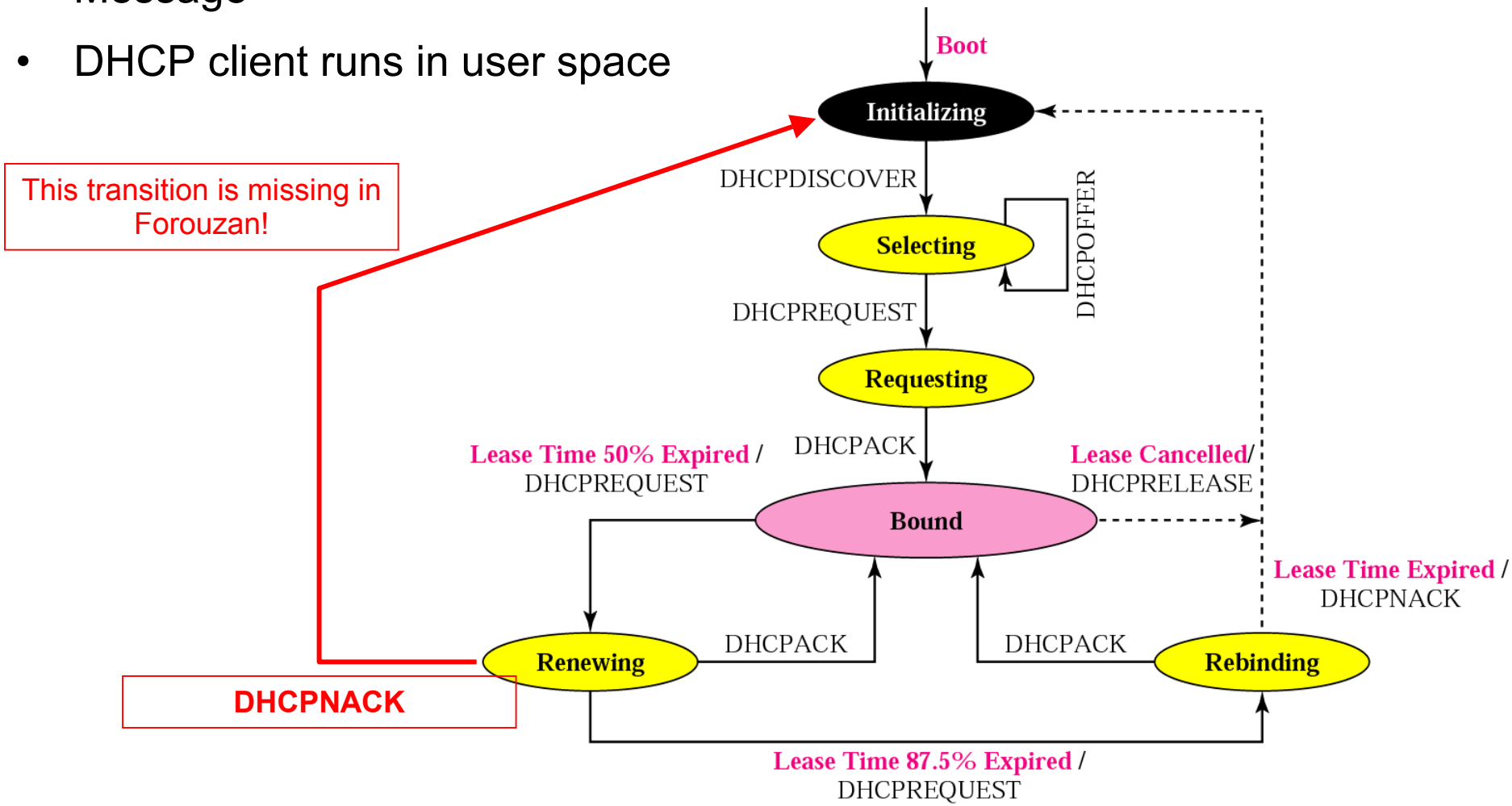
DHCP Expiration

- After 50% of lease, client need to renew
- If server NACKs, client needs to start over immediately (shown below)
- If server does not respond, client continues until lease expires



DHCP Client

- Message type option specifies DHCP Message
- DHCP client runs in user space



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DHCP's Importance

- Allows reuse of address, which avoids having to tie up addresses for systems which are not currently connected to the Internet
- Avoids user configuration of IP address (avoids mistakes and effort)
- Allows recycling of an IP address when devices are scrapped

IPv6 and ICMPv6

The Next Generation

IPv6

- Changes since IPv4 was developed (mid 70's)
  - Provider market has changed dramatically
  - Immense increase in user and traffic on the Internet
  - Rapid technology advancement
  - Bandwidth increase from kb/s to Tb/s
- IPv4 issues
  - Too few addresses (though only 3-7% of address space used)
  - Too large routing tables
- To address these issuees IETF has standardized IPv6
  - IPv6 should keep most of the characteristics of IPv4 (good design)
  - Changing the address fields is the big thing with IPv6
  - While modifying the header, improvements have been introduced

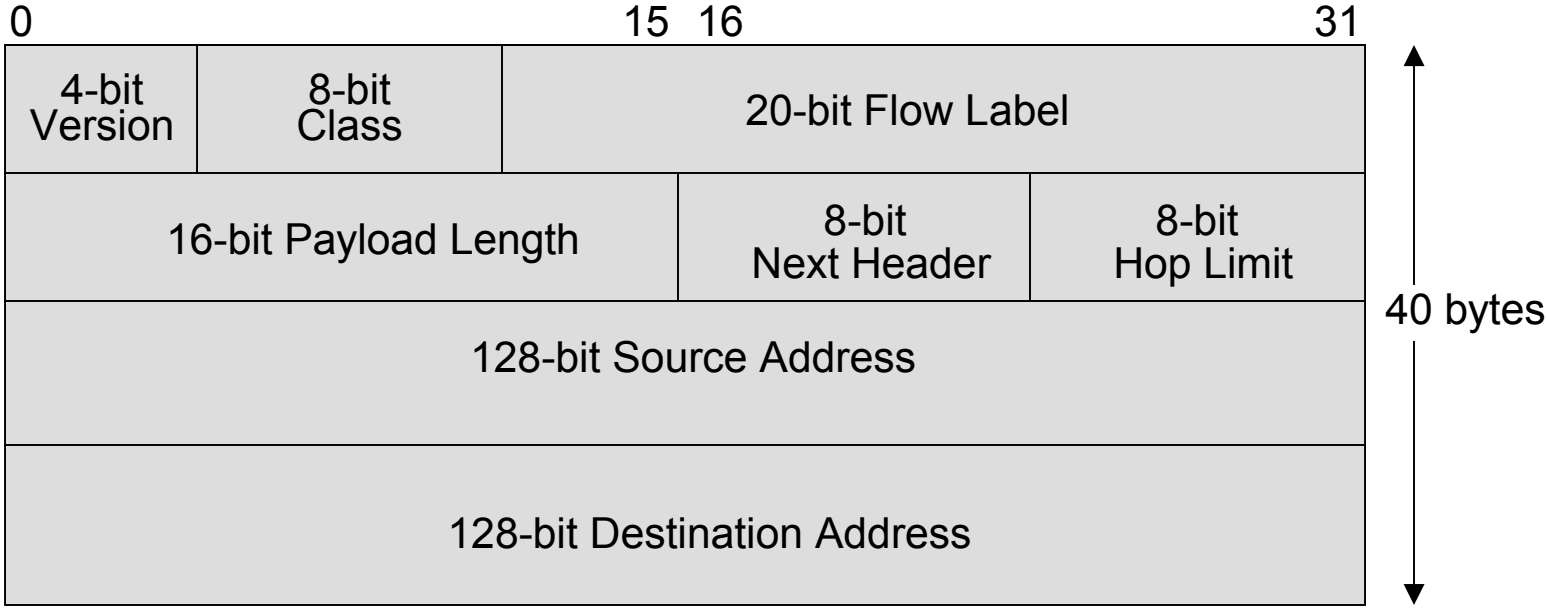
IPv6 vs IPv4

- Changes in IPv6 compared to IPv4
  - 128 bit addresses
  - extended address hierarchy
  - simplified header
  - simpler and better support for options
  - possible to extend the protocol
  - support for autoconfiguration (plug-and-play)
  - support for QoS treatment
  - host mobility
  - security
  - provider selection
  - no fragmentation in routers

IPv6 Simplifications

- Fixed format headers
  - Use extension headers instead of options
- Remove header checksum
  - Rely on link layer and higher layers to check integrity of data
- Remove hop-by-hop segmentation
  - Fragmentation only by sender due to path MTU discovery

IPv6 Header Format



- **Version** Only field identical to IPv4. Code is 6 in IPv6
- **Class** New field. Revised concept of priority bits. Facilitates handling of real-time traffic.
- **Flow Label** New field. To distinguish packets requiring the same treatment.
- **Payload Length** Replaces *length* field in IPv4. Gives length of data following IPv6 header
- **Next Header** Replaces *protocol* field in IPv4. Extension headers can be used.
- **Hop Limit** Replaces *TTL* field in IPv4. Hop limit more accurately reflects the use of TTL.
- **Src Address** Revised *source address* field. 128 bits in IPv6 vs 32 bits in IPv4.
- **Dst Address** Revised *destination address* field. 128 bits in IPv6 vs 32 bits in IPv4.

IPv4 vs IPv6 Header

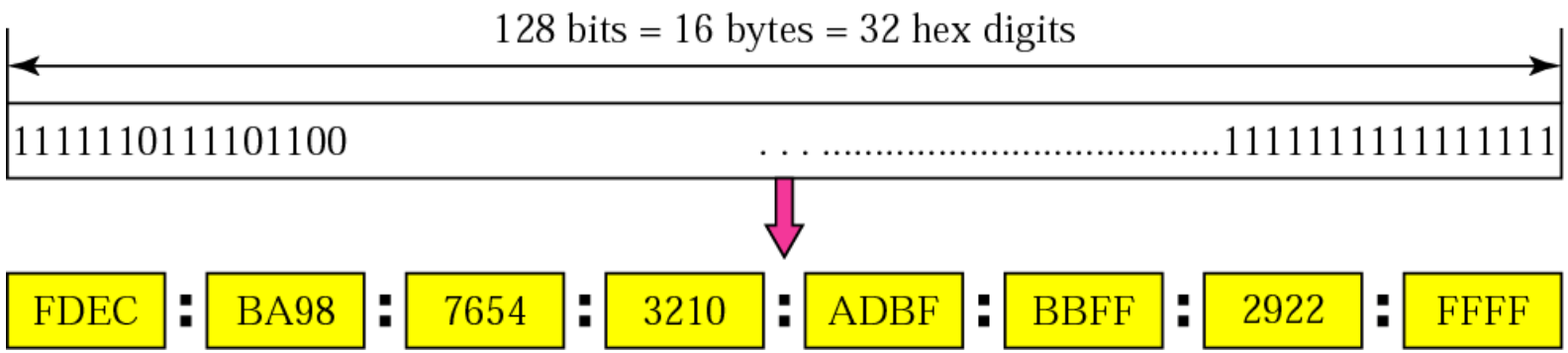
1. Header length removed
2. ToS → Class + Flow label
3. Total length →Payload Length
4. Identification, flags and offset are removed
   - Fragmentation extension header
5. TTL → Hop limit
6. Protocol → Next Header
7. Header checksum removed
8. Options → Extension headers

IPv6 Addresses

- An IPv6 unicast address identifies an interface connected to an IP subnet (as is the case in IPv4)
- One big difference between IPv6 and IPv4 is that IPv6 routinely allows each interface to be identified by several addresses
  - facilitates management
- IPv6 has three address categories:
  - unicast - identifies exactly one interface
  - multicast - identifies a group; packets get delivered to all members of the group
  - anycast - identifies a group; packets normally get delivered to nearest member of the group
- 128 bits results in 2^{128} addresses
  - Distributed over the Earth: 665,570,793,348,866,943,898,599/m^2
  - Pessimistic estimate with hierarchies: ~1,564 addresses/m^2

IPv6 address

- Colon hexadecimal notation (eight 16 bit hexadecimal integers)



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Address abbreviations and CIDR

- Leading zeros may be oppressed
  - FDEC:BA98:0074:3210:000F:BBFF:0000:FFFF →
  - FDEC:BA98:74:3210:F:BBFF:0:FFFF
- Zero compression: one of a series of zeros may be replaced by ::
  - But only once
  - FDEC:0:0:0:0:BBFF:0:FFFF →
  - FDEC::BBFF:0:FFFF
- CIDR notation to specify the first N bits of an address
  - FDEC:0:0:0:0:BBFF:0:FFFF/60

Initial Address Allocation (Oct 2002)

Prefix	Fraction of space	Allocation
0000 0000	1/256	Unassigned (some address formats)
0000 0001	1/256	Unassigned
0000 001	1/128	NSAP allocation
0000 01	1/64	Unassigned
0000 1	1/32	Unassigned
0001	1/16	Unassigned
001	1/8	Global Unicast Addresses
010	1/8	Unassigned
011	1/8	Unassigned
100	1/8	Unassigned
101	1/8	Unassigned
110	1/8	Unassigned
1110	1/16	Unassigned
1111 0	1/32	Unassigned
1111 10	1/64	Unassigned
1111 110	1/128	Unassigned
1111 1110 0	1/512	Unassigned
1111 1110 10	1/1024	Link Local Unicast Addresses
1111 1110 11	1/1024	Site Local Unicast Addresses
1111 1111	1/256	Multicast Addresses

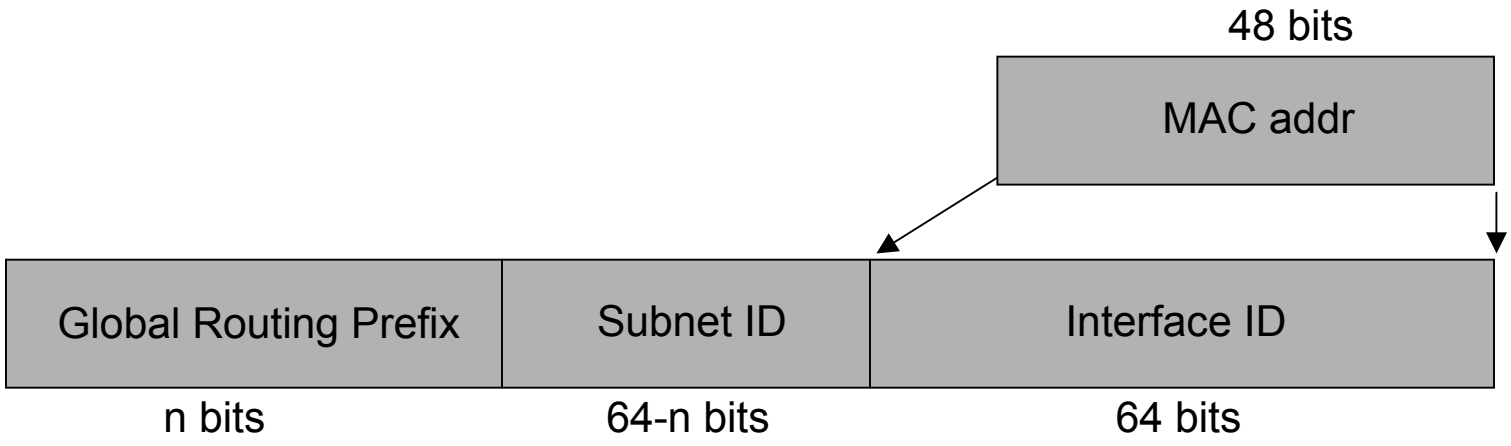
General Unicast Addresses

General format for IPv6 global unicast addresses:

Global Routing Prefix	Subnet ID	Interface ID
n bits	m bits	128-n-m bits

- Global Routing Prefix
  - typically hierarchically structured value assigned to a site (cluster of subnets/links)
- Subnet ID
  - identifier of a link within the site
- Interface ID
  - if prefix starts with binary 000: no constraint on Interface ID structure
  - otherwise: a globally unique EUI-64 address, (can be derived from 48-bit Ethernet address)

Global Unicast Addresses (001_b)



- 001_b prefix
  - 64 bit network-id (global prefix typically /32)
  - 64 bit interface-id
- MAC-derived Interface ID
  - Globally unique EUI-64 address,
  - Derived from 48-bit IEEE 802 address
  - Insert FF:FF in the middle two bytes.
  - L3→L2 address resolution automatic

Privacy Extensions RFC 3041

- MAC-derived interface-ids is an easy way to ensure unique addresses
  - And get rid of L2/L3 address resolution
- But, you know many things about the origin of the packet
  - IEEE 802 addresses have encoded data
    - The identity and vendor of the interface card
    - You can derive which equipment you use
    - E.g., exploit bugs in that equipment
  - You can track the node when netid is changed (dhcp/mobile IP)
- Mac-derived interface ids
  - Randomly assigned interface-id
  - Changes over time
- Protects users
- You need to make L2/L3 address resolution

Special Address Formats

The following special addresses are allocated from prefix 0000 0000:

- Unspecified Address (0:0:0:0:0:0:0:0:0)
  - only used as source address during bootstrap by a computer that has not yet learned its address
- Loopback Address (0:0:0:0:0:0:0:0:1)
  - used for testing software (compare with IPv4 loopback address 127.0.0.1)
- IPv6 Addresses with Embedded IPv4 Addresses
  - Needed during transition from IPv4 to IPv6 (checksum calculation unaffected)

0000.....0000	0000	IPv4 Address
0000.....0000	FFFF	IPv4 Address
80 bits	16 bits	32 bits

0000 – *IPv4-compatible IPv6 address*: to dynamically tunnel IPv6 packets over IPv4 routing infrastructure

FFFF – *IPv4-mapped IPv6 address*: to represent the address of IPv4 nodes as IPv6 addresses

Link-local unicast address

- Link-Local addresses - for use on a single link
  - For purposes such as automatic address configuration, neighbour discovery, isolated network
- Routers do not forward packets using link-local addresses

1111111010	0000...0000	Interface ID
10 bits	54 bits	64 bits

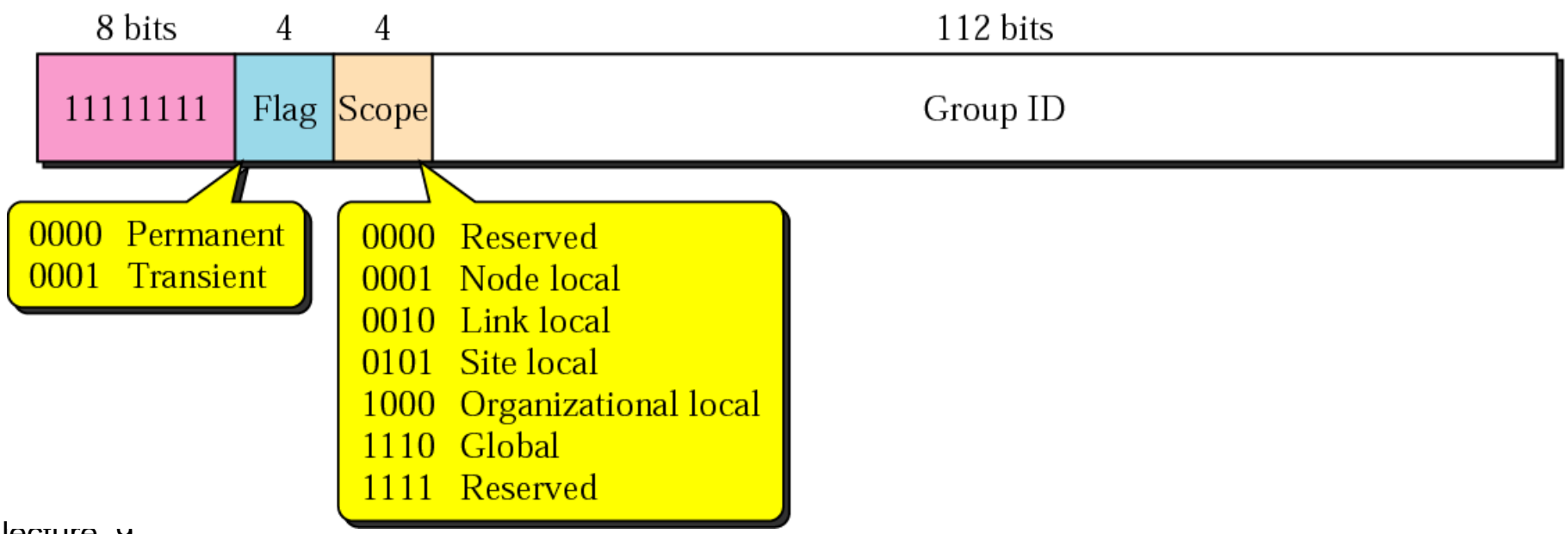
Site-local unicast address

- Site-local addresses – for use on a single site
  - For purposes such as private or nonroutable addressing
- Routers do not forward site-local addresses outside the site
- Now obsolete

1111111011	Subnet ID	Interface ID
10 bits	54 bits	64 bits

Multicast Addresses

- Permanent: assigned by IANA
- Scope Example: assume NTP servers have a group ID of 101:
  - FF02::101 means all NTP servers on the same link as the sender
  - FF05::101 means all NTP servers in the same site as the sender
  - FF0E::101 means all NTP servers in the Internet



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Predefined Multicast Addresses

Some predefined mcast addresses:

- All nodes multicast
  - FF01::1 (interface-local)
  - FF02::1 (link-local)
  - In IPv4 224.0.0.1 is used
- All routers multicast
  - FF01::2 (interface-local)
  - FF02::2 (link-local)
  - FF05::2 (site-local)
  - In IPv4 224.0.0.2 is used

Anycast Addresses

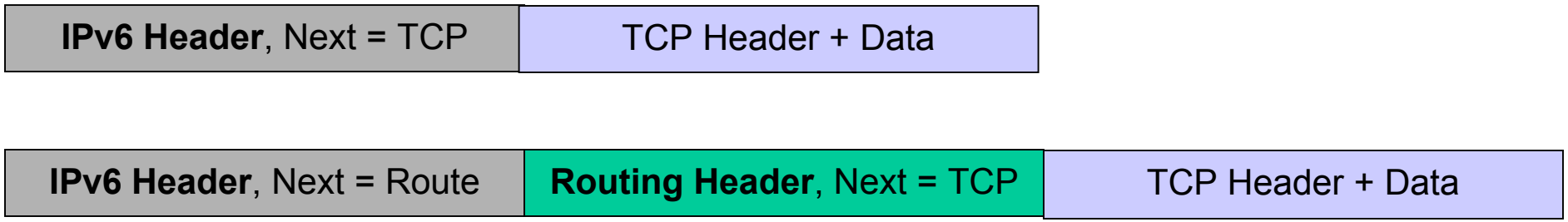
- Sending a packet to a generic address to get a specific service from the "nearest" instance. This puts the burden of determining which instance to deliver it to on the routing system.
- According to IPv6 Addressing Architecture Draft
  - An anycast address is an address assigned to more than one interface (typically different nodes)
  - Anycast addresses are allocated from the unicast address space
  - Nodes must be explicitly configured to know an address is an anycast address
- Expected use:
  - identify the set of routers belonging to a service provider
  - identify the set of routers providing an entry into a particular routing domain
- Restrictions (until more experience has been gained):
  - An anycast address must *not* be used as source address of an IPv6 packet
  - An anycast address must *not* be assigned to an IPv6 host, only to an IPv6 router

Current IPv6 address allocation

- IANA + Regional Internet Registries (RIR) only allocates 001_2 addresses
- Currently (2003) 480 prefixes allocated
- 94% of these prefixes are /32.
- From current 6bone BGP routing table statistics, it can be noted that only 2% of the prefixes are longer than /48 [7].

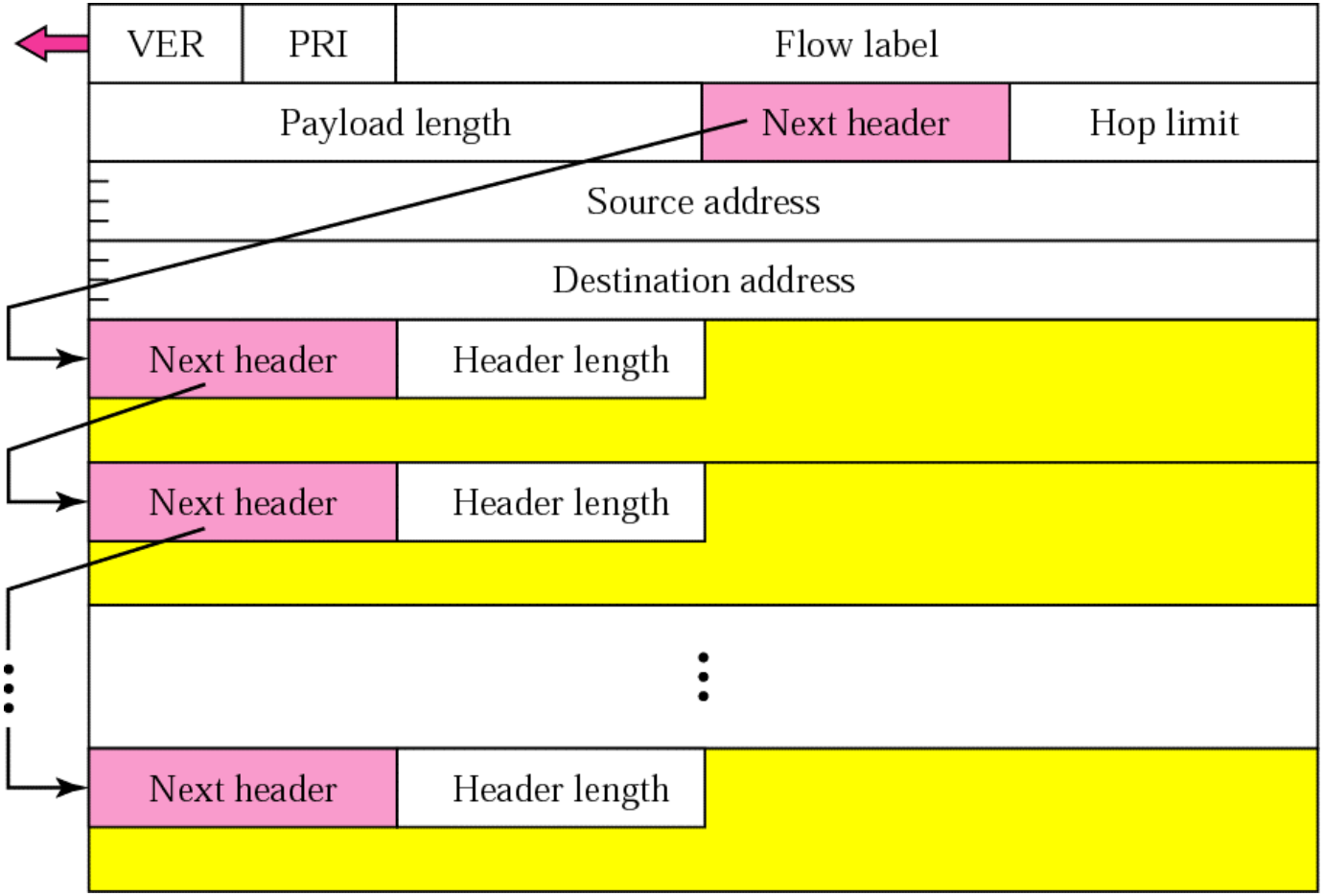
Extension Headers

- To give more functionality to IP, extension headers have been introduced in IPv6
- Several of the IPv6 extension headers are options in IPv4
- Extension headers are placed between the IPv6 base header and the transport level header (TCP/UDP)



Extension headers are linked

- Next headers come in a linked list
- Can be hard to parse: need an iterative process



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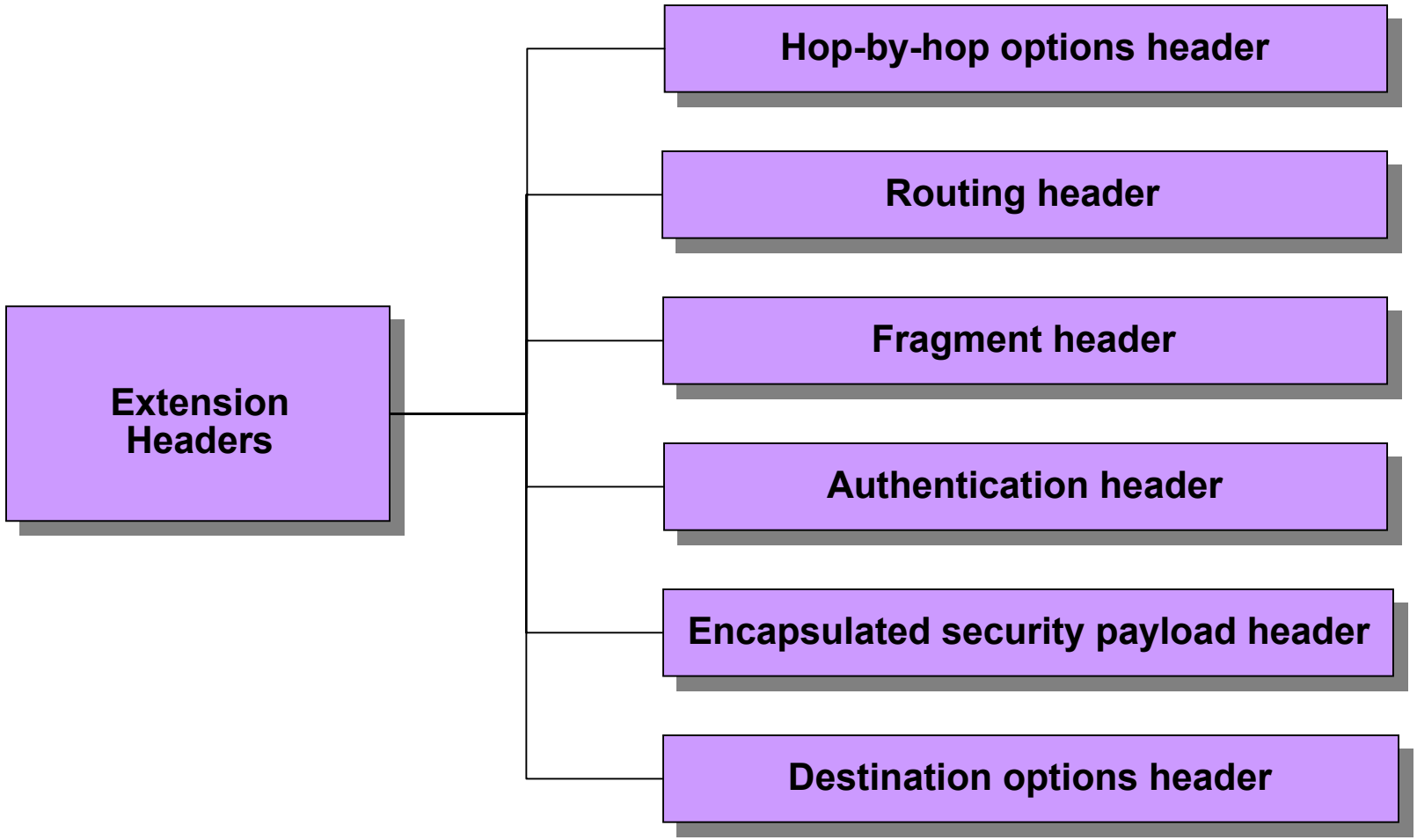
Extension header types



Extension Headers

- Hop-by-hop Options header
  - TLV coded options processed by every hop along the path
  - Jumbo payload option for pkt > 65535 bytes (RFC 2675)
  - Router alert option (RFC 2711)
- Fragment header
  - Only source can fragment packets in IPv6
  - Source must use Path MTU Discovery (RFC 1981) or send max 536 bytes payload
  - Fragmentation information
    - Fragmentation offset shifted (as in IPv4)
    - Fragmentation ID is 32 bits (16 bits in IPv4)
    - No DF flag (present in IPv4, but not needed in IPv6)

Extension headers, cont'd

- Routing header
  - Strict or loose source routing
  - Similar to the IPv4 Source route and Record route options
- Authentication header (IPSEC)
  - to validate the message sender and ensure integrity of data
- Encapsulated Security Payload header (IPSEC)
  - to provide confidentiality and guard against eavesdropping
- Destination Options header
  - TLV coded options processed by destination only

Autoconfiguration (Plug-and-Play)

- Address resolution
  - ICMP has been revised along with the development of IPv4 → IPv6
  - IPv6 does not use ARP but a neighbour detection scheme based on ICMPv6
- Stateful configuration (managed)
  - Flag in *router advertisement* tells whether to rely on autoconfiguration or to use conventional managed configuration (DHCP)
- Stateless autoconfiguration / Serverless

Stateless autoconfiguration

- Use link-local address and interface ID
- Hosts join all-nodes mcast address (FF02::1)
- Hosts communicate to routers using all-routers mcast address (FF02::2)
- ICMPv6 *router solicitation* sent by host to request additional information
- ICMPv6 *router advertisement* sent by router to inform host about prefixes for site and global addresses

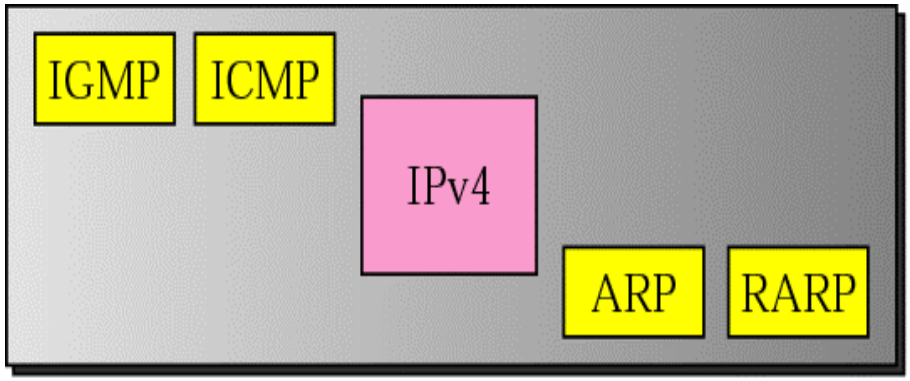
Real-Time Support and Flows

8-bit Class	20-bit Flow ID

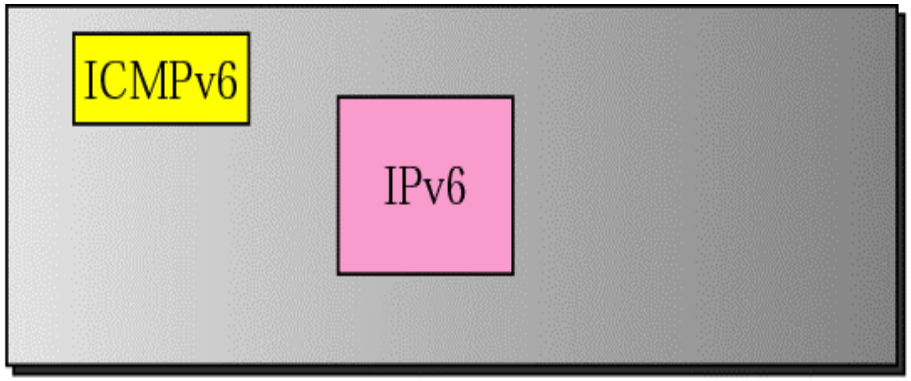
- **Flow ID field:** used by a source to label sequences of packets for which it requests special handling by the IPv6 routers
- **Class field:** available for use by originating nodes and/or forwarding routers to identify and distinguish between different classes or priorities of IPv6 packets
- The use of these fields is still experimental, and subject to change as the requirements become clearer
- Flow ID assigned to a flow by the flow's source node
- All packets belonging to the same flow must be sent with the same src addr, dst addr, and flow label
- RSVP or other mechanism needed for resource reservation
- Real-time data transfers require protocols such as RTP in addition to IPv6

Network Layer Comparison - v4 vs v6

- ICMPv4 has been modified to be more suitable for IPv6, and thus updated to ICMPv6
- ARP and IGMP in version 4 are now part of ICMPv6
- RARP has been dropped due to limited use (DHCP does the job of RARP)
- As in ICMPv4, ICMPv6 messages are divided into 2 categories:
  - Error-reporting (somewhat different messages in v6 vs v4, see following slide)
  - Query (rather different messages in v6 vs v4, see following slide)



Network layer in version 4



Network layer in version 6

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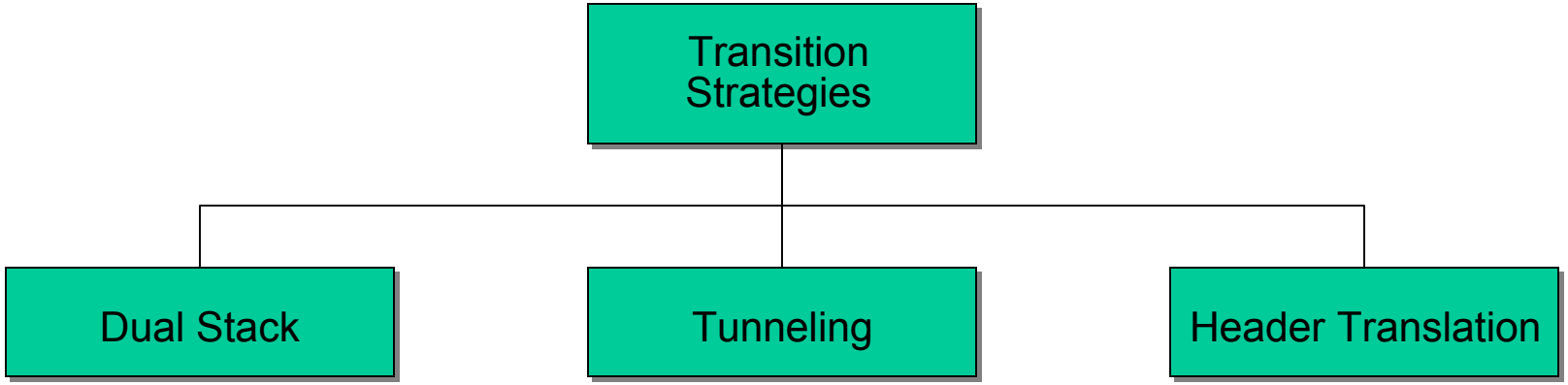
ICMPv4 vs ICMPv6

Error Report Message – Type	Ver 4	Ver 6
Destination unreachable	Yes	Yes
Source quench	Yes	No
Packet too big	No	Yes
Time exceeded	Yes	Yes
Parameter problem	Yes	Yes
Redirection	Yes	Yes

Query Message – Type	Ver 4	Ver 6
Echo request and reply	Yes	Yes
Timestamp request and reply	Yes	No
Address mask request and reply	Yes	No
Router solicitation and advertisement	Yes	Yes
Neighbour solicitation and advertisement	ARP	Yes
Group membership	IGMP	Yes

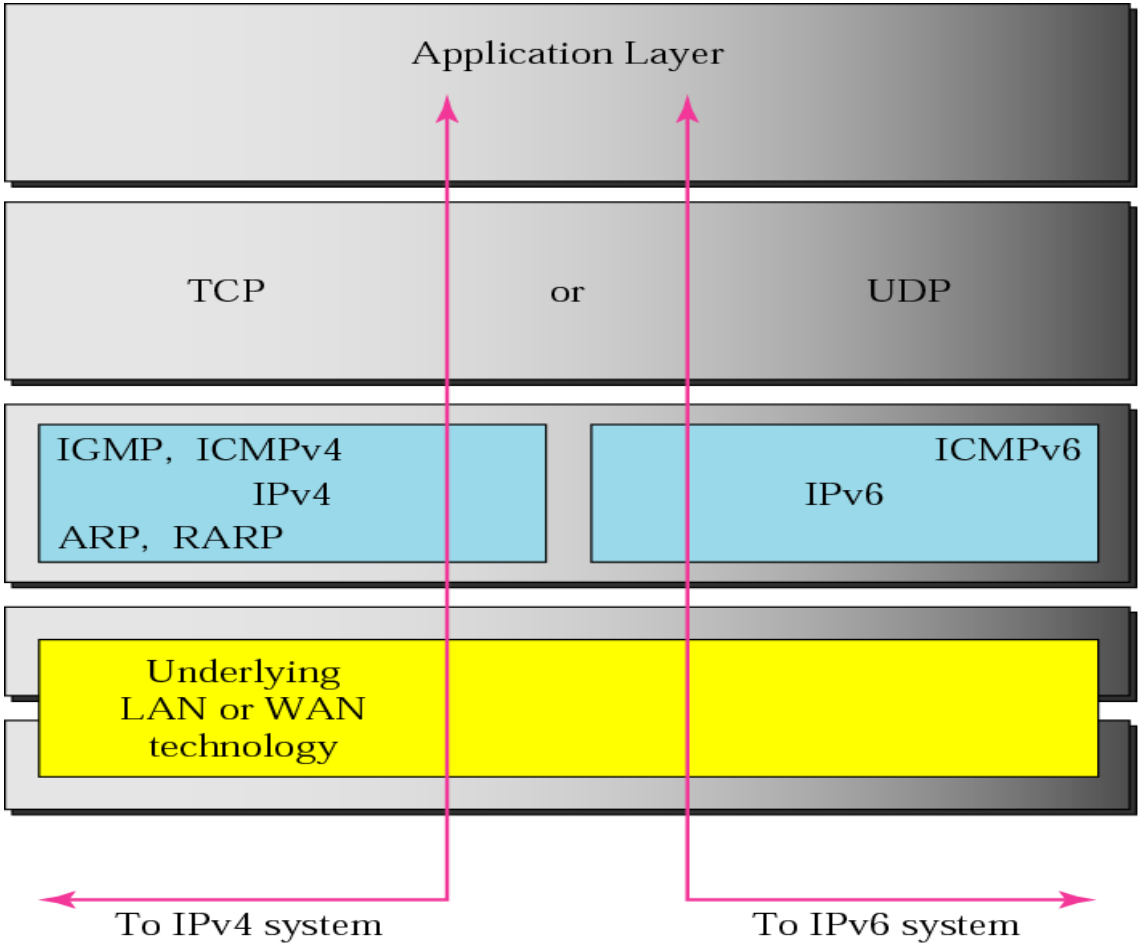
Transition from IPv4 to IPv6

- Because of the large number of systems on the Internet, the transition from IPv4 to IPv6 cannot happen suddenly
- Transition should be smooth to prevent problems
- Three transition strategies have been devised by IETF



Dual Stack

- All hosts have dual stack of protocols until all of the Internet runs IPv6
- To determine which version to use, the source host queries the DNS



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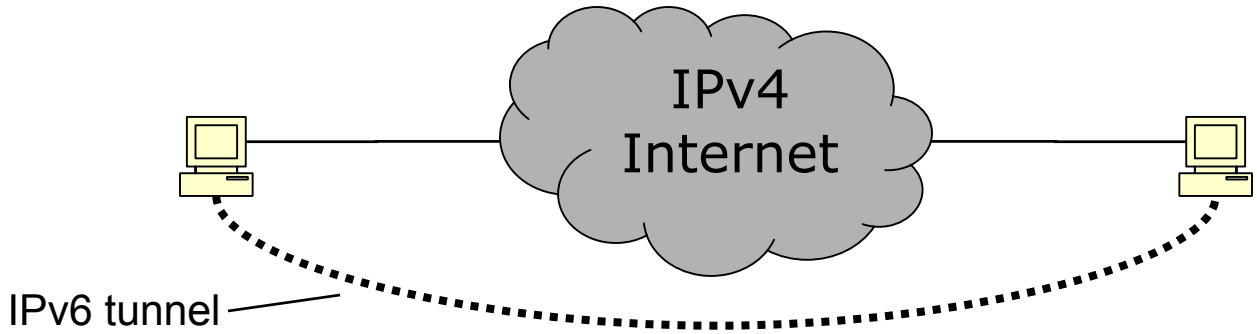
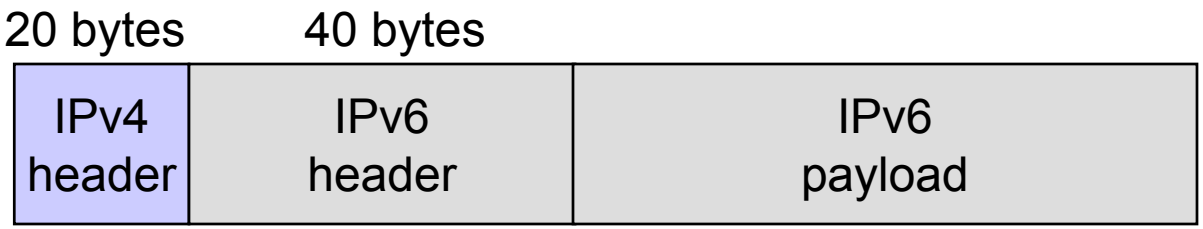
Transition from IPv4 to IPv6



- Tunnel appear as a virtual link between IPv6 nodes
- Encapsulation of IPv6 packets in IPv4



IPv6 Summary

- IPv6 has:
  - 128-bit address space
  - revised header format
  - new options
  - allowance for extension
  - support for special handling of packet flows
  - increased security measures
- IPv6 uses hexadecimal colon notation with abbreviation methods
- IPv6 has three address types: unicast, anycast, and multicast
- IPv4, ICMPv4, ARP, RARP, and IGMP replaced with IPv6 and ICMPv6
- IPv4 to IPv6 transition strategies are based on dual-stack and tunneling